Math Door to Door: The Traveling Salesperson Problem

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The Traveling Salesperson Problem (TSP) is a well known NP-hard problem (Rodríguez-Almazán et al., 2026, Sec. 1) highlighting the computational limits of computing and the frontier of computational computer research. This problem is approachable when creating a set limit or to find a "good enough answer" but becomes difficult when adding more stops to the salesperson route. The table below shows that at 10 cities to stop at there are 181,440 distinct tours assuming symmetric distances (Rich, 2007, Sec. 27.1). As the problem scales factorially the problem becomes harder to solve until there would realistically be a problem too large to solve within a reasonable amount of time (Rodríguez-Almazán et al., 2026, Sec. 1). Recently with the rapid development of new computing abilities, researchers have taken new approaches in solving this problem utilizing different techniques and approaches (Rodríguez-Almazán et al., 2026; Sun and Yang, 2023; Xia et al., 2024). This paper will evaluate these new approaches to the problem based on their systematic setup and the mathematics it uses. Each section will detail improvement in approach based on mathematical or computational advancements made since the problem was posed.

$$N(n) = \frac{(n-1)!}{2}, \qquad n \geq 3, \qquad d(u,v) = d(v,u). \tag{1}$$

Table 1: Distinct tours with symmetric distances (Rich, 2007, Sec. 27.1).

Cities n	Distinct tours $N(n)$
4	3
5	12
10	181,440
15	43,589,145,600
20	60,822,550,204,416,000

The first approach in evolution of this problem is the brute force approach (Rich, 2007, Sec. 27.1). This approach tests each possible route to check if it is the shortest one. The conceptual approach is to first generate a route. The next step is to calculate total length. Then you compare the route's total length to other routes' total length and keep the shortest route. By checking each possible route the approach determines the shortest possible route by keeping the shortest possible route and getting rid of the others. This approach gives an exact answer because of evaluating each one (Rich, 2007, Sec. 27.1). The formula (2) shows how r, which is the route, and how route length can be calculated (Rich, 2007, Sec. 27.1).

$$\begin{aligned} r &= (v_1, v_2, \ldots, v_n, v_1), \\ L(r) &= \sum_{i=1}^{n} d(v_i, v_{i+1}), \qquad v_{n+1} = v_1. \end{aligned} \tag{2}$$

$d(v_i, v_{i+1})$ this part of the formula calculates the distance from one city to another. Then $v_{n+1} = v_1$ ensures the return to the starting city. This formula checks each route's total length verifying that the best route is the exact shortest route. This approach fails as the number of routes grows; the factorial growth of possible routes becomes significantly difficult to solve using this approach (Rich, 2007, Sec. 27.1). This has led to a shift in mindset to find approaches that reduce the solution space and can still be examined.

Next, academic and research advancements in possible solutions to TSP came in introducing new approaches to reduce the search space during summing an answer (Rodríguez-Almazán et al., 2026, Sec. 2). Instead of checking every route, researchers introduced pruning partial branches calculated to have a longer total route length than the best known routes; a computer can eliminate route trees that would produce longer route results, therefore saving search space (Rodríguez-Almazán et al., 2026, Sec. 2). Another approach used was developing heuristics that can allow for solutions that are approximate but reasonable in the time used to search for an answer (Rodríguez-Almazán et al., 2026, Sec. 2). These two approaches highlight development in advancing the solutions to the TSP problem by changing approaches and mindsets to find answers. These approaches also limit the search time needed to solve this type of problem, an important signal of progress in providing solutions to these problems.

In 2026, Rodríguez-Almazán et al. began a different approach to tackle the TSP problem (Rodríguez-Almazán et al., 2026). When using a single route the order is one route at a time assuming brute force is used. Classical computers using classical algorithms were also restricted to these specific approaches when using algorithms to reduce search space. Researchers introduced using quantum computing to differentiate this approach from past attempts at solving the problem (Rodríguez-Almazán et al., 2026, Sec. 1). These researchers used Quantum Phase Estimation and also Grover-style amplitude amplification to search for better routes (Rodríguez-Almazán et al., 2026, Secs. 3–5). This conceptualizes the problem and changes the mathematics of the search. Over time as technology has advanced so has the approach to solve this problem. This draws into question: is the solution just a matter of time, or will this problem continue to elude researchers and mathematicians?

From brute force to utilizing quantum mechanics, the Traveling Salesperson Problem high-

lights the timeline and how approaches have evolved over time. Since the first recording of this problem researchers from around the world have approached this famously difficult NP-hard problem with different mindsets and approaches to a solution (University of Waterloo, 2022; Rodríguez-Almazán et al., 2026). Using brute force ensures an exact answer, but fails in respect to valuing search time and real-world constraints (Rich, 2007, Sec. 27.1). These limiting factors lead to researchers developing approaches such as pruning partial branches or using a heuristics approach to shrink the search time needed (Rodríguez-Almazán et al., 2026, Sec. 2). The use of quantum mechanics highlights the evolving nature of how researchers are approaching the problem and how new developments in computing ability could unlock new approaches and solutions to the problem (Rodríguez-Almazán et al., 2026, Sec. 7).

References

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